Spectral Geometries in Three Dimensions

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November 11, 2026

Abstract

Every three-dimensional manifold that is locally homogeneous, i.e. looks the same everywhere, can be classified into one of eight model geometries. We explore the extent to which the type of a manifold can be recovered from its Laplace eigenvalues, and which model geometries admit pairs of manifolds that are isospectral but not isometric

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Draft: November 11, 2026

STATUS: Sections 1–3 are essentially complete while Section 4 is outlined and Section 5 contains the plan of attack for a research problem. Sections 2–3 are background, Section 4 exposits Lin-Schmidt-Sutton's theorem that the Thurston geometry of a closed locally homogeneous 3-manifold can be determined from its spectrum, with expected new work on distinguishing Sol *and* $\widetilde{SL_2}(\mathbb{R})$ *which is the one case that they left open. Section 5 will aim to prove a new result on the audibility of Berger lens spaces.*

1 Introduction

Spectral geometry is the study of the relationship between the geometry of a space and the eigenvalues of the Laplacian on the space. The Laplacian is a differential operator that encodes how heat spreads and waves propagate on a space, and spectral geometry is intimately connected with the study of partial differential equations, dynamics, differential geometry, and many other fields.

Spectral geometry began with Mark Kac's question, "Can one hear the shape of a drum?" [Kac66]. Through the wave equation, the eigenvalues $\{\lambda_k\}$ of the Laplacian Δ of a planar domain Ω can be interpreted as the resonant frequencies of a drum in the shape of Ω. Though Kac asked this question for planar domains, or more generally domain $\Omega \subset \mathbb{R}^n$, it can easily be formulated for an arbitrary Riemannian manifold (M, g) using the Laplace-Beltrami operator Δ_g and its eigenvalues. For this version, Milnor [Mil64] found 16-dimensional tori with the same Laplace spectrum which are not isometric. After some more examples of such manifolds (called *isospectral* manifolds), Sunada [Sun85] found a general method to produce many examples of isospectral but non-isometric Riemannian manifolds. In the following years, Brooks, Bérard, Buser, and many others steadily refined the Sunada method[1] until Gordon-Webb-Wolpert in [GWW92] finally produced an example of isospectral but non-isometric 2D drums.

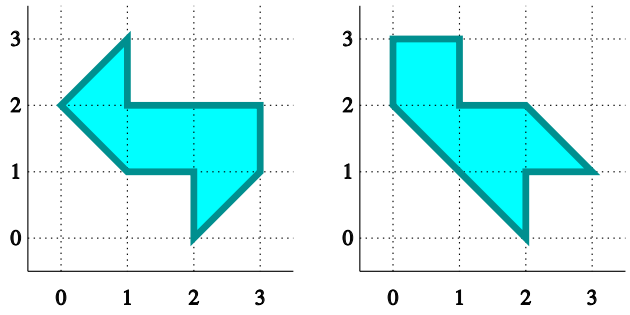


Figure 1: Gordon-Webb-Wolpert's isospectral drums

Because of the existence of isospectral non-isometric manifolds, we cannot hope to determine the shape of a drum from its spectrum. We can instead ask, "How much of the shape of a drum can you hear"? Another way of overcoming this problem is to restrict attention to certain sub-classes of spaces and ask whether we could distinguish two manifolds belonging to those spaces. In three dimensions, a very natural class of manifolds to look at are the compact *locally homogeneous manifolds*. They are the spaces that "look the same everywhere." Moreover, just like how every compact 2-dimensional manifold can be classified into a Euclidean, elliptic, or hyperbolic manifold, each 3-dimensional manifold can be cut up into pieces that admit locally homogeneous metrics that can be classified into *eight* different geometries, called *model geometries*. Because of this, locally homogeneous spaces form natural building blocks for geometry.

Given that we are exploring this class of manifolds, there are some natural questions we may ask about them:

1. Can locally homogeneous manifolds be distinguished from non locally-homogeneous manifolds via their

[1] An account of the improvement of Sunada's method from the original to Gordon, Webb, and Wolpert's method is given in [Ara22]

spectra, or can a locally homogeneous manifold have the same spectrum as a non locally-homogeneous manifold?

2. Do two isospectral locally homogeneous manifolds necessarily have the same model geometry?

3. Do two isospectral locally homogeneous manifolds necessarily have the same Riemannian universal cover?

4. Are two isospectral locally homogeneous manifolds necessarily isometric?

If we can hear local homogeneity, comparing locally homogeneous manifolds only to other locally homogeneous manifolds does not miss any isospectral pairs. Restricting to locally homogeneous manifolds, the second question asks whether the spectrum can tell us which of the eight model geometries our manifold corresponds to. In two dimensions, this is true since a compact locally homogeneous two-dimensional manifold has constant sectional curvature and we can distinguish these using the heat invariants (see Section 2.3).

In three dimensions, we already have non-isometric isospectral manifolds. Thus, we investigate the weaker versions of the claim, asking whether isospectral locally homogeneous manifolds have the same universal cover, or the same model geometry. We do this in Section 4, expositing the result in [LSS24].

Different model geometries behave very differently. We can already see this in two dimensions: two isospectral round spheres must be isometric, as must be two isospectral flat tori. However, isospectral hyperbolic surfaces need not be isometric. Thus in two dimensions the model geometry $\mathbb{H}^2$ behaves very differently from $\mathbb{E}^2$ or S^2. In three dimensions, Euclidean geometry $\mathbb{E}^3$ behaves very well, with isospectral manifolds being isometric (with one exceptional pair, "tetra" and "didi"; see [DR04]), and the hyperbolic geometries $\mathbb{H}^3$ and $\mathbb{H}^2 \times \mathbb{R}$ are similarly wild, with many non-isometric isospectral examples. We will explain these differences in detail. Unlike the two-dimensional case, spherical geometry is not as well understood in dimension three. In Section 5 we will explain the status of these elliptic manifolds, and as a new result rule out a candidate isospectral pair.

Conventions. All manifolds are closed, connected, and orientable unless stated otherwise. We will use the geometer's Laplacian $\Delta = -\operatorname{div} \circ \operatorname{grad}$. If integrating over a manifold, especially if integrating over one factor of a product manifold, we will use $\int_M f(x)dx$ to mean integrating against the Riemannian volume form on the manifold from which x is drawn. We will use the Riemannian curvature convention $R(X,Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$. If a property (such as being a space form) or quantity (such as volume) is deducible from the Laplace spectrum, then we will say that it is 'audible' or that we can 'hear' it, or that it is 'spectrally determined'.

Outline of this thesis. In section 2, we define the basic objects of spectral geometry and give an account of the standard results. Then, in section 3 we explain geometric structures on manifolds and enumerate the geometries we will be working with. In section 4 we aim to show that a 3-dimensional geometric manifold's geometry is audible, using invariants inspired by the heat equation. In section 5, we aim to use invariants inspired by the wave equation to prove certain lens spaces are *not* isospectral.

2 Background

2.1 The Laplace Operator

The Laplace operator is involved in many important partial differential equations. On $\mathbb{R}^n$, it is defined as[2]

$$\Delta = -\left(\frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2}\right)$$

Suppose we have a domain $\Omega \subset \mathbb{R}^n$, and the temperature at every point is given by a function $u(x,t)\colon \Omega \times \mathbb{R}_{>0} \to \mathbb{R}$. Then the heat propagates according to the *heat equation*

$$\Delta u + \frac{\partial}{\partial t} u = 0$$

(where we have set constants depending on the material to be 1). If we attach Ω to a heat bath kept at a fixed temperature along $\partial\Omega$ given by a function $f\colon \partial\Omega \to \mathbb{R}$, then we get the *Dirichlet boundary condition* $u|_{\partial\Omega} = f$, and the steady-state solution would be given by a u for which $\Delta u = 0$. We could instead require $\partial\Omega$ to be a perfect insulator, letting no heat in or out. In this case we would get the *Neumann boundary conditions* $\frac{\partial u}{\partial \mathbf{n}} = 0$ where $\mathbf{n}$ is the outward unit normal on the boundary (in other words there is no heat flow along the perpendicularly out direction), and the steady state solution would still be given by $\Delta u = 0$ with Neumann boundary conditions. In fact this latter case gives us just constants (on a connected compact domain), which agrees with out intuition that heat spreads out and equalizes over time.

The Laplacian is also involved in wave propagation. Think of Ω being the surface of a drum or a pond, and $u(x,t)$ being the height of the water/material at a given point x at time t. Then u satisfies the *wave equation*

$$\frac{\partial^2 u}{\partial t^2} + \Delta u = 0$$

In this case, Dirichlet boundary conditions mean that the boundary is fixed and cannot vibrate (such as a drum), while Neumann boundary conditions imply that the boundary is free, like a string that is free to vibrate on one end, or water against the wall of a pond.

Now suppose $u(x,t)$ is the complex-valued wavefunction of a particle living on Ω (with no external force). Then the probability that the particle is found in some region $X \subset \Omega$ is given by $\frac{\int_X |u(x,t)|^2 dx}{\int_\Omega |u(x,t)|^2 dx}$. This wavefunction evolved according to the *Schrödinger equation*

$$i\frac{\partial u}{\partial t} = \Delta u$$

We will focus mainly on the heat and wave equations. Consider the wave equation $u_{tt} + \Delta u = 0$ with $u|_{\partial u} = 0$, and use the ansatz[3] $u(t,x) = f(t)g(x)$. Then $u_{tt}(t,x) = f''(t)g(x)$ and $\Delta u = f(t)\Delta g$, so $f''(t)g(x) + f(t)\Delta g = 0 \implies \frac{f''(t)}{f(t)} = \frac{-\Delta g}{g(x)}$. The right-hand side of this equation depends only on x and the left-hand side only on t, so they must both be equal to some constant $-\lambda$. The solutions to $f''(t) = -\lambda f(t)$ is given by $a\cos(\sqrt{\lambda}t) + b\sin(\sqrt{\lambda}t)$, while the solutions to $\Delta g = \lambda g$ are precisely the Laplace eigenfunctions. Thus, the eigenfunctions and eigenvalues of the Laplacian play a key role in solving these partial differential equations.

It can be shown that the set $\{\lambda \in \mathbb{R} \mid \Delta g = \lambda g\}$ for appropriate bounded domains $\Omega \subset \mathbb{R}^n$ is discrete, and a subset of $\mathbb{R}_{\geq 0}$ and that the eigenspace for each eigenvalue is a finite-dimensional subspace of $L^2(\Omega)$; see [LMP23,

[2] this sign convention is the *geometer's Laplacian*, which gives us positive eigenvalues and the equation $\Delta = d\delta + \delta d$

[3] to be fully rigorous we have to later justify that the solutions we get from this ansatz are all of them, via functional analysis

Chapter 2]. Thus, we can index the eigenvalues as $\lambda_1 \le \lambda_2 \le \cdots$, repeating per multiplicity, and then eigenfunctions are φ_k satisfying $\Delta\varphi_k = \lambda_k\varphi_k$. The eigenvalues accumulate only at ∞ and $\lambda_k \to \infty$ as $k \to \infty$. The eigenfunctions can be chosen to form an orthonormal basis of $L^2(\Omega)$.

For any of these above equations, we can apply a similar separation-of-variable technique as we did for the wave equation. With appropriate boundary conditions we get that an arbitrary solution is given by [Can16]

$$u(x,t) = \sum_{k\ge 1} a_k(t)\varphi_k(x)$$

where φ_k is the Laplacian eigenfunction as given above, and (for constants a_k and b_k)

$$a_k = \begin{cases} a_k e^{-\lambda_k t} & \text{for the heat equation} \\ a_k e^{-i\sqrt{\lambda_k}t} + b_k e^{i\sqrt{\lambda_k}t} & \text{for the wave equation} \\ a_k e^{-i\lambda_k t} & \text{for the Schrödinger equation} \end{cases}$$

(note that $a_k e^{-i\sqrt{\lambda_k}t} + b_k e^{i\sqrt{\lambda_k}t}$ generalizes our previous solution since $e^{it} = \cos t + i\sin t$, and we recover the previous solution when the imaginary parts cancel out).

We can ask these same questions on arbitrary Riemannian manifolds, and the equations (and general solution techniques) remain the same, except that the Laplacian is replaced by the *Laplace-Beltrami* operator. This generalization of the Laplacian to Riemannian manifolds uses the fact that $\Delta = -\operatorname{div}\circ\operatorname{grad}$ on $\mathbb{R}^n$. To define it, we need to define a divergence and gradient on arbitrary manifolds. The standard coordinate expressions on $\mathbb{R}^n$ are not invariant under coordinate transformations, and in fact these operators depend on the chosen Riemannian metric g.

Suppose we have a Riemannian manifold (M,g). The gradient $\operatorname{grad} f$ of a function f is the vector field representing the 1-form df. That is, for a vector field X, $g(\operatorname{grad} f, X) = df(X) = Xf$. Sometimes this is written as $(df)^\sharp$ using the 'raising operator' $\sharp$. This gradient satisfies all the standard properties of the gradient in $\mathbb{R}^n$, such as being orthogonal to level sets (if N is a level set of f, then for any $v \in T_pN$, $vf = 0$, thus $g((\operatorname{grad} f)_p, v) = df(v) = vf = 0$). In coordinates, it is given by

$$\operatorname{grad} f = \sum_{i=1}^n \sum_{j=1}^n g^{ij}\frac{\partial f}{\partial x^i}\frac{\partial}{\partial x^j}$$

The divergence of a vector field measures how much it is a 'source' or a 'sink.' A Riemannian manifold has a canonical volume form dV_g; in a local coordinate frame it is given by $\sqrt{\det g}\,dx^1 \wedge \cdots \wedge dx^n$. The divergence of a vector field X is the unique function satisfying $\mathcal{L}_X(dV_g) = (\operatorname{div} X)dV_g$. In this sense the divergence precisely measures how much the volume form changes along the flow of the vector field. Using Cartan's formula for Lie derivatives $\mathcal{L}_X\omega = d(\iota_X\omega) + \iota_X(d\omega)$ (where ι_X is the 'interior derivative' sending an n-form ω to the $(n-1)$-form $\omega(X,\cdot)$), we get the formula $d(\iota_X dV_g) = (\operatorname{div} X)dV_g$. Thus, in coordinates,

$$\operatorname{div} X = \frac{1}{\sqrt{\det g}}\sum_{i=1}^n \frac{\partial}{\partial x^i}(X^i\sqrt{\det g})$$

Finally, the Laplacian can be defined as $\Delta_g = -\operatorname{div}\circ\operatorname{grad}$. Its expression in coordinates is given by

$$\Delta f = -\frac{1}{\sqrt{\det g}}\sum_{i=1}^n\sum_{j=1}^n \frac{\partial}{\partial x^i}\left(g^{ij}\frac{\partial f}{\partial x^j}\sqrt{\det g}\right)$$

The coordinate-invariant definition has the benefit that it makes obvious the fact that for any isometry $\psi\colon M \to M$, $\psi^*\circ\Delta = \Delta\circ\psi^*$. This is because both the gradient and divergence operators commute with isometries.

Proposition 2.1. If ψ is any diffeomorphism of N then $\psi^*\circ\Delta_g = \Delta_{\psi^* g}\circ\psi^*$. Thus, if ψ is an isometry ($\psi^* g = g$), the Laplacian commutes with it

The coordinate-based expression tells us that the Laplacian is an elliptic differential operator, and by the regularity results on elliptic operators on compact manifolds (or manifolds with boundary, with appropriate boundary conditions) along with Green's theorem, we get the same results about its eigenvalues: they are nonnegative, discrete, and go to infinity, and Laplacian eigenfunctions can be chosen to given an orthonormal basis of $L^2(M)$.

Remark 2.2. There is another definition of the Laplacian on compcat oriented manifolds, which generalizes beyond functions. The inner product g induces inner products on all the exterior powers $\Lambda^k T^*M$ and by integrating these we get an inner product on the space of sections $\Omega^k(M)$ given by $\langle \alpha, \beta \rangle = \int_M \langle \alpha_p, \beta_p \rangle_p dV_g$. We can consider the space $\Omega^\bullet(M) = \bigoplus_{k=0}^n \Omega^k(M)$, and we then get an inner product on this space by declaring the different degrees orthogonal. The operator $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$ has a formal adjoint δ which decreases degree, and satisfies $\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle$. Using this property we can show that for a 1-form ω, $\delta\omega = -\operatorname{div}(\omega^\sharp)$. Then the Laplacian can be defined as $\Delta = \delta d$ To construct this explicitly, we need the Hodge star $* : \Omega^k(M) \to \Omega^{n-k}(M)$ defined by $\langle \alpha, \beta \rangle = \int_M \alpha \wedge *\beta$, and defined on an orthonormal coframe $\{\varepsilon^i\}$ by sending ε^I to $\operatorname{sign}(I, [n] \setminus I)\varepsilon^{[n]\setminus I}$ (so for instance, $*\varepsilon^1 = \varepsilon^2 \wedge \cdots \wedge \varepsilon^n$ and $*f = f dV_g$). Then, $\delta = (-1)^{nk+n+1} * d *$. and we can define the *Laplacian on forms* as a map $\Delta_k : \Omega^k(M) \to \Omega^k(M)$ by $d\delta + \delta d = (d+\delta)^2$. This matches the definition for functions since $\delta f = 0$ for any function. One reason for defining it like this is that $*\Delta f = d\delta(*f)$, so this is the 'right' definition on n-forms. The actual reason for this definition comes from the *Hodge theorem* on closed oriented manifolds: there is an isomorphism $\ker \Delta_k \cong H^k_{dR}(M)$ given by sending a harmonic form (a form ω with $\Delta\omega = 0$) to its cohomology class, so each cohomology class is represented by exactly one harmonic form.

2.2 Isospectral manifolds

With these definitions we can return to Kac's [Kac66] question on hearing the shape of a drum. Given a bounded domain in $\mathbb{R}^n$, or more generally a compact Riemannian manifold (M, g), we get a sequence $\lambda_k \nearrow \infty$ of eigenvalues. These represent the 'natural frequences of vibration,' and their eigenfunctions form an orthonormal basis of L^2. If we are given that two spaces have the same eigenvalues, are they the same shape?

Definition 2.3. Two Riemannian manifolds (M, g) with the same Laplacian eigenvalues are said to be *isospectral*. If there is a boundary, then we can specify whether they are *Dirichlet isospectral* or *Neumann isospectral* based on the boundary conditions applied.

Thus, we can tell the shape of a drum if isospectral manifolds are also isometric. Otherwise, we would never be able to distinguish between isospectral but non-isometric manifolds from their spectrum alone.

Remark 2.4. Using the Laplacian on forms defined in Remark 2.2, we get a sequence of Laplace eigenvalues λ^d_k for d-forms $\Omega^d(M)$. Two manifolds that have the same Laplace eigenvalues not just on functions but also on forms are called *strongly isospectral*, and for all the questions we have asked for isospectral manifolds, we can also ask them for strongly isospectral manifolds. In this thesis we shall restrict to the Laplacian on functions.

What information can we recover from the spectrum? For a start, we can recover the volume, so isospectral drums must have the same size. This result is called *Weyl's law* and is generally proven using variational methods (for instance in [LMP23]). We will give a proof using the heat trace in Section 2.3.

Theorem 2.5 (Weyl's law, [LMP23, Theorem 3.3.4])**.** Let (M, g) be an n-dimensional manifold. If it has a nonempty boundary, we can impose either Dirichlet or Neumann conditions. Let $N(\lambda)$ be the number of eigenvalues of M less than or equal to λ. Then,

$$\lim_{\lambda \to \infty} \frac{N_k(\lambda)}{\lambda^{\frac{n}{2}}} = C_n \operatorname{Vol}(M)$$

where C_n is a constant depending only on the dimension n

For manifolds with boundary, using heat trace asymptotics, we will see that we can also recover $\operatorname{Vol}(\partial M)$. If we consider the case of planar domains $\Omega \subset \mathbb{R}^2$ with piecewise-linear or smooth boundary, Kac's analysis in

[Kac66] gives us some more precise results, such as hearing the number of holes in the domain. We will return to some of this in Section 2.3

2.2.1 Flat tori

The first example of drums whose shape couldn't be heard came shortly after Kac asked his question. Milnor in [Mil64] constructed two flat sixteen-dimensional tori which were isospectral but not isometric. To explain this construction, we will follow the exposition of [NRR22].

Definition 2.6. A *flat torus* is a torus $S^1 \times \cdots \times S^1$ equipped with a metric of constant sectional curvature zero.

Proposition 2.7 ([NRR22, §2.1])**.** It can be shown that flat tori are quotients $\mathbb{R}^n/\Gamma$ where $\mathbb{R}^n$ is equipped with the Euclidean metric and Γ is a full-rank lattice i.e. an additive subgroup. Each such lattice is given by $A\mathbb{Z}$ where $A \in O(n)$. Two flat tori $\mathbb{R}^n/\Gamma_1$ and $\mathbb{R}^n/\Gamma_2$ are isometric if and only if $\Gamma_1 = X\Gamma_2$ for $X \in O(n)$. If $\Gamma_1 = A_1\mathbb{Z}$ and $\Gamma_2 = A_2\mathbb{Z}$ then this occurs if and only if $CA_1 = A_2B$ for some $B \in \mathrm{GL}_n(\mathbb{Z})$ and $C \in O(n)$. The volume of $\mathbb{R}^n/\Gamma$ for $\Gamma = A\mathbb{Z}$ is given by $\det(A)$ which we will denote $\mathrm{Vol}(\Gamma)$

Thus, the isometry class of a flat torus is determined exactly by the *congruence class* of the lattice it is given by. Moreover, the spectrum is also determined by the lattice.

Definition 2.8. Let $\Gamma \subset \mathbb{R}^n$ be a lattice. Then the *dual lattice* Γ^* is defined by

$$\Gamma^* = \left\{ l \in \mathbb{R}^n \mid \langle l, \gamma \rangle \in \mathbb{Z} \right\}$$

This is itself a lattice, and if $\Gamma = A\mathbb{Z}$ then $\Gamma^* = (A^{-1})^T\mathbb{Z}$

Proposition 2.9 ([NRR22, Theorem 2.8])**.** The Laplace spectrum of a flat torus $\mathbb{R}^n/\Gamma$ is

$$\left\{ 4\pi^2 ||l||^2 \mid l \in \Gamma^* \right\}$$

and and the eigenfunction associated to $l \in \Gamma^*$ is given by $\varphi_l(x) = e^{2\pi i \langle x, l \rangle}$

Sketch of proof. The map $\mathbb{R}^n \to \mathbb{R}^n/\Gamma$ is a Riemannian covering map, and it can be shown (e.g. [LMP23, §6.2]) that Laplace eigenfunctions on the quotient are precisely the (images of) deck-invariant eigenfunctions on the Riemannian cover. On $\mathbb{R}^n$, which is not compact, Δ has a continuous spectrum, with eigenfunctions $\varphi_y(x) = e^{2\pi i \langle x, y \rangle}$ for all $y \in \mathbb{R}^n$, whose eigenvalue is $||y||^2$. The deck-invariant eigenfunctions are precisely those for which

$$\begin{aligned} & e^{2\pi i \langle x+\gamma, y \rangle} = e^{2\pi i \langle x, y \rangle} \forall \gamma \in \Gamma \\ \iff & e^{2\pi i \langle \gamma, y \rangle} = 1 \\ \iff & \langle \gamma, y \rangle \in \mathbb{Z} \\ \iff & y \in \Gamma^* \end{aligned}$$

To make this proof precise, we would have to deal with the theory of the Laplacian on noncompact manifolds and justify using $e^{2\pi i \langle x, y \rangle}$ even though those are not in L^2 □

Thus both the spectrum and the geometry of a flat torus is controlled entirely by algebraic properties of the lattice defining it. Because everything can be calculated explicitly, flat tori give us an easy case to test and demonstrate general results. One of these is the connection between the Laplace spectrum and the *length spectrum*.

Definition 2.10. For a closed Riemannian manifold (M, g), the *length spectrum* $\mathrm{Spec}_L(M)$ is the set of lengths of closed geodesics.

Remark 2.11. If the manifold instead has a boundary, the length spectrum is a little harder to define, because we have to answer questions like: what happens when a geodesic hits the boundary? What happens if it hits a corner? Upon hitting a boundary, it normally reflects, but if it is tangent to the boundary it can 'glide' along. Different cases are studied under the names 'billiards' and 'diffraction.' These are all relevant to the 2D drums case, but since we will focus on closed Riemannian manifolds we will not go into the different ways of defining the length spectrum in all these other cases. For a brief overview, see [HLR17, §3] or the introduction of [GM79].

Since a flat torus is a quotient of $\mathbb{R}^n$, whose geodesics are straight lines, geodesics on $\mathbb{R}^n/\Gamma$ are also straight lines. Looking at them on the universal cover, a straight line from the origin is closed in the quotient if and only if it ends up at one of the points of Γ. Thus,

Proposition 2.12. The set of closed geodesics of a flat torus $\mathbb{R}^n/\Gamma$ is $\{\gamma \in \Gamma\}$ and $\mathrm{Spec}_L(\mathbb{R}^n/\Gamma) = \{||\gamma|| \ \forall \gamma \in \gamma\}$

The length spectrum and the Laplace spectrum of a flat torus are connected by the *Poisson summation formula*

Theorem 2.13 (Poisson summation formula, [NRR22, Theorem 2.12])**.** For a full-rank lattice $\Gamma \subset \mathbb{R}^n$, the following series converge for any $t \in (0,\infty)$, and satisfy

$$\sum_{\gamma^* \in \Gamma^*} e^{-4\pi^2 ||\gamma^*||^2 t} = \frac{\mathrm{Vol}(\Gamma)}{(4\pi t)^{\frac{n}{2}}} \sum_{\gamma \in \Gamma} e^{-||\gamma||^2/4t}$$

The sum $\sum_{\gamma^* \in \Gamma^*} e^{-4\pi^2||\gamma^*||^2 t}$ is the *heat trace* $\sum_{k\geq 0} e^{-\lambda_k t}$. Another important sum connects flat tori to modular forms

Definition 2.14. The *theta series* of a lattice Γ is given by

$$\theta_\Gamma(z) = \sum_{\gamma \in \Gamma} e^{i\pi z ||\gamma||^2}$$

where $z \in \mathbb{C}$ with $\mathrm{Im}(z) > 0$

Theorem 2.15 ([NRR22, Corollary 2.15])**.** Let two flat tori be given by lattices Γ and Λ. The following are equivalent

1. $\mathbb{R}^n/\Gamma$ and $\mathbb{R}^n/\Lambda$ are isospectral
2. $\mathbb{R}^n/\Gamma$ and $\mathbb{R}^n/\Lambda$ have the same heat traces
3. $\mathbb{R}^n/\Gamma$ and $\mathbb{R}^n/\Lambda$ have the same length spectrum (counted with multiplicities)
4. Γ and Λ have the same θ-series

Proof. Isospectrality implies having the same heat trace, and having the same length spectrum implies having the same theta series. For both of them, going the other way requires the same technique, so we demonstrate with the heat trace. We will show that if we have $f(t) = \sum_{\gamma^* \in \Gamma^*} e^{-4\pi^2||\gamma^*||^2 t} = \sum_{k\geq 0} e^{-\lambda_k t}$, then we can recover each λ_k. First, write $\sum_{k\geq 1} e^{-\lambda_k t} = \sum_{i \geq 1} m_i e^{-\lambda_i t}$ where each λ_i is unique and has multiplicity m_i. Then, for $\lambda < \lambda_1$, $\lim_{t\to\infty} e^{-\lambda t}(f(t)-1) = 0$ while $\lim_{t\to\infty} e^{-\lambda_1 t}(f(t)-1) = m_1$. Then, define $f_1(t)$ to be $f(t) - 1 - m_1 e^{-\lambda_1 t}$, and repeat. Inductively this process determines all the λ. To apply this on the theta series, use $z = it$.

By the Poisson summation formula, two manifolds have the same heat trace if and only if the have the same value of $\sum_{\gamma\in\Gamma} e^{-||\gamma||^2/4t}$ for every t; by applying the same inductive argument and letting t go to 0^+, we can recover the length spectrum from $\sum_{\gamma\in\Gamma} e^{-||\gamma||^2/4t}$. The theorem follows. □

With this setup, we can explain Milnor's construction in [Mil64]. First, he cites two sixteen dimensional lattices (E_{16} and $E_8 \times E_8$) which are not congruent. By [NRR22, Theorem 3.5], in even dimensions, the θ series of a lattice is a modular form. In sixteen dimensions, there is only one modular form for $\mathrm{PSL}_2(\mathbb{Z})$, so the theta series of both these lattices is the same. Thus, we have found isospectral non-isometric tori.

The state-of-the-art for isospectral tori is detailed in [NRR22]. It is not hard to show that two flat two-dimensional isospectral tori must be isometric (see discussion under Theorem 5.2 in [NRR22]). Schiemann [Sch97] proved the same for three-dimensional tori; his result is stated in the language of quadratic forms, because each lattice determines a quadratic form through its matrix, and asking if two lattices are congruent is the same as asking if two integral quadratic forms are integrally equivalent. In dimension four, Schiemann found an isospectral pair, demonstrating that three is the largest dimension where we can hear the shape of a flat torus.

2.2.2 Sunada's method

Following Milnor's tori, there were sporadic counterexamples using different techniques specific to their cases. Then, Sunada [Sun85] came up with a general method to produce isospectral manifolds. Sunada's method took as input group-theoretic data and produced from it isospectral non-isometric manifolds. The group theoretic data is the following:

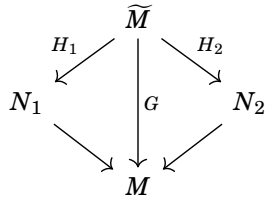
Definition 2.16. A *Sunada triple* or (*Gassmann-Sunada triple*) is a triple (G, H_1, H_2) where G is a finite group, H_1, H_2 subgroups of G, and H_1 and H_2 are not conjugate in G, but intersect each conjugacy class of G in the same number of elements (this called being *almost conjugate*). That is,

$$\#(H_1 \cap [g]) = \#(H_2 \cap [g]) \ \ \forall g \in G$$

where $[g] = \{g' \in G \mid g' = hgh^{-1}, h \in G\}$

Theorem 2.17 (Sunada's theorem)**.** Let (G, H_1, H_2) be a Sunada triple and M be a closed manifold such that there exists a surjection $\varphi\colon \pi_1(M) \twoheadrightarrow G$. Let $\widetilde{M}$ be the cover whose deck group is G (so $M = \widetilde{M}/G$), and let N_1 and N_2 be the intermediate covers given by $N_i = \widetilde{M}/H_i$. For any metric g on M let $\tilde{g}$, g_1, and g_2 be the pullback metrics on $\widetilde{M}$, N_1, and N_2. Then, (N_1, g_1) is isospectral to (N_2, g_2). Further, g can be chosen so that (N_1, g_1) is not isometric to (N_2, g_2)

The situation is shown below. To make matters simpler, suppose $\pi_1(M) = G$. Since every finite group is the fundamental group of a compact 4-manifold, we can find a manifold satisfying this for every Sunada triple (G, H_1, H_2). Then, we get a covering diagram as below. Eigenfunctions on M are the G-invariant eigenfunctions on $\widetilde{M}$, and eigenfunctions on N_i are the H_i-invariant eigenfunctions on $\widetilde{M}$.



Before we prove this theorem, let us explain some moving parts. In the setup above, almost-conjugacy of H_1 and H_2 is the condition that gives us isospectrality. However, if H_1 and H_2 are conjugate, then the conjugation map between them gives us an isometry between $\widetilde{M}/H_1$ and $\widetilde{M}/H_2$. Thus, we need the condition that H_1 and H_2 are almost-conjugate but not conjugate. This still doesn't rule out an isometry between $\widetilde{M}/H_1$ and $\widetilde{M}/H_2$; to do this, we choose a metric such that the only isometries of $\widetilde{M}$ come from G. Then, an isometry between N_1 and N_2 would have to come from a deck transformation on $\widetilde{M}$, which would conjugate H_1 to H_2. Thus, the Sunada condition blocks (N_1, g_1) and (N_2, g_2) from being isometric. In fact, there are examples where H_1 and

H_2 are almost-conjugate but not even isomorphic as groups, in which case (if $\pi_1(M) = G$) N_1 and N_2 are not even homeomorphic, let alone isometric.

Sunada considered his method "a geometric analogue of a routine method in number theory." The original application of this technique was by Gassmann in [Gas26] to find number fields that were not isomorphic but had identical Dedekind zeta functions, and used the properties of Galois correspondence between Galois groups and field extensions. Sunada's method is essentially the same idea applied to the Galois correspondence between fundamental groups and covering spaces. Because Gassmann was the first to consider the applications of almost-conjugate but not conjugate subgroups, Sunada triples are often called Gassmann-Sunada triples.

There are many proofs of Sunada's theorem, and [Bro99] gives an overview of four different ones. We will sketch the ones using the heat trace and the length spectrum in Section 2.3. In this section we give an account of the representation-theoretic proof of Sunada's method, which relies on the fact that Laplace eigenspaces are representations of G and the eigenspaces of N_i are (isomorphic to) the H_i-invariant subspaces of these G-representations.

Definition 2.18. A representation (V, ρ) of a group G is a vector space (often complex) V together with a linear left-action ρ of G on V. This is equivalent to a homomorphism $\rho : G \to GL(V)$. If $H \leq G$ then V^H is the subspace of H-invariant vectors i.e $V^H = \{v \in V \mid \rho(h) \cdot v = v \text{ for all } h \in H\}$

Proposition 2.19. Let $E_\lambda(M)$ be the subspace of $L^2(M)$ consisting of λ-eigenfunctions of the Laplacian. Then $E_\lambda(M)$ is a representation $\mathrm{Isom}(M, g)$ by setting $\rho(\psi) \cdot f = (\psi^*)^{-1}(f)$.

Proof. Recall from Proposition 2.1 that the Laplacian commutes with isometries i.e. $\psi^* \circ \Delta = \Delta \circ \psi^*$. Let $f \in E_\lambda(M)$ so $\Delta f = \lambda f$, and let $\psi \in \mathrm{Isom}(M, g)$. Then

$$\Delta(\psi^* f) = \psi^*(\Delta f) = \psi^*(\lambda f) = \lambda \psi^* f$$

where the last equality follows from linearity of pullback. Thus $\psi^* f \in E_\lambda(M)$.

Now, let ψ be an isometry; then ψ^{-1} is also an isometry, so if $f \in E_\lambda(M)$ so is $(\psi^{-1})^* f = \rho(\psi) \cdot f$. Since the pullback is a linear map on functions, the isometry group acts linearly on $E_\lambda(M)$; the axioms of a group action follows from $\mathrm{Id}^* f = f$ and $(\varphi \circ \psi)^*(f) = (\psi^* \circ \varphi^*)(f)$ (this contravariance is what led us to define the action via an inverse). □

Proposition 2.20. Let $\pi : M \to M/H$ be a Riemannian covering. Then every Laplace eigenvalue of M/H is also an eigenvalue of M, and the multiplicity of λ on M/H is the dimension of the H-invariant subspace of $E_\lambda(M)$. In other words, $\dim E_\lambda(M/H) = \dim E_\lambda(M)^H$.

Proof. Let $f \in E_\lambda(M/H)$ and $\widetilde{f} = \pi^* f$ i.e. $\widetilde{f}(x) = f(\pi(x))$. Then f^* is H-invariant. Now, consider an evenly-covered open set $U \subset M/H$ and a V covering it so that $\pi|_V : V \to U$ is an isometry (this we can do since covering maps are local isometries). Then, $\widetilde{f}|_V = \pi^* f_U$, and $\Delta_V \pi^* f_U = \pi^* \Delta_U f|_U = \pi^* \lambda f_U = \lambda \widetilde{f}_V$. Thus $\widetilde{f}_V$ is a λ-eigenfunction on the open set V. Cover M with such open sets. Since $\Delta \widetilde{f}$ and $\lambda \widetilde{f}$ are equal on every set of the cover they are equal on M. Thus, λ is also an eigenvalue of M.

Functions on M/H are in bijection with H-invariant functions on M, and by the previous paragraph, λ-eigenfunctions on M/H correspond to H-invariant λ-eigenfunctions on M. Thus, $\dim E_\lambda(M/H) = \dim E_\lambda(M)^H$. □

Lemma 2.21. Let (V, ρ) be a finite-dimensional representation of a group G and $H \leq G$ be a finite subgroup. Then

$$\dim(V^H) = \mathrm{Trace}\left(\frac{1}{|H|} \sum_{h \in H} \rho(h)\right) = \frac{1}{|H|} \sum_{[g]} \#([g] \cap H) \mathrm{Trace}(\rho(g))$$

where $[g]$ represents the conjugacy class of $[g]$ and the sum on the right ranges over the conjugacy classes in G.

Proof. Let $\psi = \frac{1}{|H|}\sum_{h\in H}\rho(h)$. We will show that ψ is a projection onto V^H. Then, for $v \in V^H$, $\psi(v) = v$, so $\psi|_{V^H}$ is the identity. Now, we want to show that the image of ψ is in V^H. Choose $h_0 \in H$ and consider

$$\begin{aligned}\rho(h_0)\cdot\psi(v) &= \frac{1}{|H|}\sum_{h\in H}\rho(h_0)\rho(h)v \\ &= \frac{1}{|H|}\sum_{h\in H}\rho(h_0h)v \\ &= \frac{1}{|H|}\sum_{h'\in H}\rho(h')v \\ &= \psi(v)\end{aligned}$$

where we have used the fact that left-multiplication by h_0 simply rearranges the order in which we iterate through the elements of H. Thus, $\psi(V)\subset V^H$. Since $\psi|_{V^H} = \mathrm{Id}_{V^H}$, $\psi^(v) = \psi(\underbrace{\psi(v)}_{\in V^H}) = \psi(v)$, so $\psi^2 = \psi$. Thus ψ is projection onto V^H.

The dimension of the image of a projection operator is its trace, so $\dim(V^H) = \mathrm{Trace}(\psi) = \frac{1}{|H|}\sum_{h\in H}\mathrm{Trace}\,\rho(h)$. Trace is invariant under conjugation, so for every $h_1, h_2 \in [g]$ for some $g \in G$, $\mathrm{Trace}(\rho(h_1)) = \mathrm{Trace}(\rho(h_2)) = \mathrm{Trace}(\rho(g))$. Grouping the sum by conjugation classes thus turns it into $\frac{1}{|H|}\sum_{[g]}\#([g]\cap H)\mathrm{Trace}(\rho(g))$. □

Remark 2.22. Using some more representation theory, we can see that the condition of almost-conjugacy is equivalent to $\mathbf{Ind}_{H_1}^G \mathbf{1} = \mathbf{Ind}_{H_2}^G \mathbf{1}$, and then by Frobenius reciprocity we get that

$$V^H = \mathrm{Hom}_H(\mathbf{1}, \mathbf{Res}_H^G V) = \mathrm{Hom}_G(\mathbf{Ind}_H^G \mathbf{1}, V)$$

showing that $\dim V^{H_i}$ is the same for both.

Proof of Theorem 2.17. Let (G, H_1, H_2) be a Sunada triple and let $M, \widetilde{M}, N_1, N_2$ be as in the theorem statement. Choose any metric on M and pull it back to the covers. First we will show that N_1 and N_2 are isospectral. Let λ be any Laplace eigenvalue of N_1 and m be its multiplicity; by definition, $m = \dim(E_\lambda(\widetilde{M}/H_1))$. By Proposition 2.20, $m = \dim(E_\lambda(\widetilde{M})^{H_1})$, and by 2.21, this is $\frac{1}{|H_1|}\sum_{[g]}\#([g]\cap H_1)\mathrm{Trace}(\rho(g))$. Since H_1 and H_2 are almost conjugate, $\#([g]\cap H_1)\mathrm{Trace}(\rho(g)) = \#([g]\cap H_2)\mathrm{Trace}(\rho(g))$ for all conjugacy classes $[g]$. Also, since $|H_i| = \sum_{[g]}\#([g]\cap H_i)$, $|H_1| = |H_2|$. Hence these two sums are the same, and $\dim(E_\lambda(\widetilde{M})^{H_1}) = \dim(E_\lambda(\widetilde{M})^{H_2})$. Thus λ is also an eigenvalue of N_2 with the same multiplicity. This argument can be run in either direction to show that N_1 and N_2 have the same multiset of eigenvalues and are hence isospectral.

Now, choose a nowhere locally homogeneous metric g on M, i.e. a metric such that for any distinct open $U, V \subset M$, there is no isometry between $(U, g|_U)$ and $(V, g|_V)$. In [Sun85, §3], Sunada shows that non only do such metrics exist on any closed manifold, they form a residual set in the space of metrics. If g is such a metric, then the only isometries of $(\widetilde{M}, \widetilde{g})$ are in the deck group G. Now suppose $\varphi\colon N_1 \to N_2$ was an isometry. Then its lift to $\widetilde{M}$ is an isometry of $\widetilde{M}$, so must be in G, in which case $\varphi H_1 \varphi^{-1} = H_2$, contradicting the hypothesis that H_1 and H_2 are not conjugate in G. □

All of the preceding steps carry over exactly to the Laplacian on forms (Remark 2.2), so manifolds the produced isospectral manifolds are also strongly isospectral. One of the major drawbacks to this method is that it only guarantees *some* metric on the base manifold for which the coverings are non-isometric; in general it doesn't rule out non-isometry. Also, since the isospectral manifolds produced by Sunada's method are coverings of the same space, they are always locally isometric, and they cannot be simply-connected. To overcome this, Sutton in [Sut02] considers the case where G is a Lie group, with H_1 and H_2 Lie subgroups. He replaces the almost-conjugate condition with a representation-theoretic condition that is equivalent to almost-conjugacy for finite groups. By using more careful representation theory, and looking at homogeneous spaces instead of just covers, he constructs simply-connected isospectral spaces that are not locally isometric.

2.2.3 Isospectral drums in $\mathbb{R}^2$

The raw version of Sunada's method could not be used to produce isospectral drums in $\mathbb{R}^2$, so Kac's question was left open. However, the basic idea that group-theoretic input can be used to produce isospectral spaces was expanded upon after Sunada's original publication. For instance, Buser shows in [Bus92, Chapter 12] that a Sunada triple can be used to glue together pieces of a hyperboic surface in such a way as to produce isospectral surfaces. This was a very different method for producing isospectrality compared to Vignéras' [Vig80], who used arithmetic geometry.

One fruitful idea from this approach is that of transplantation. First, one takes some pieces, and glues them together in two different ways according to the two different subgroups in a Sunada triple. Then, one takes an eigenfunction on of the resulting glued spaces, and 'transplants' it onto the other piece. The gluing data ensures that the transplanted function is still an eigenfunction, giving us isospectrality. This was the method used in Gordon–Webb–Wolpert's result of isospectral but not isometric drums [GWW92] (see Figure 2). For a detailed explanation of the transplantation method, together with the history of its development from the Sunada method, see [Ara22]. The article [BCDS94] gives many examples of planar isospectral domains, and all of them are obtained by applying the data of Sunada triples.

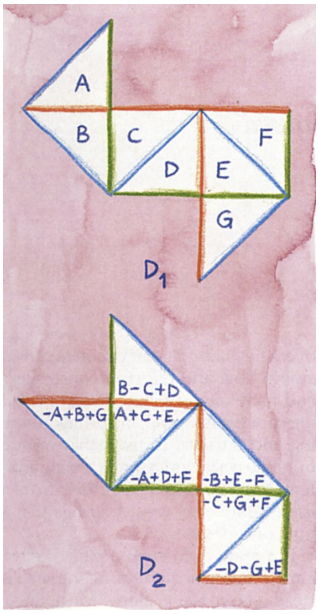


Figure 2: The transplantation data in Gordon-Webb-Wolpert's example. Figure from [GW96]

Even though Gordon–Webb–Wolpert's counterexample answered Mark Kac's question in the negative, their drums have corners and are not convex. Kac's problem is still open for smooth drums, and for convex drums.

Open Problem 2.23. If $\Omega_1, \Omega_2 \subset \mathbb{R}^2$ are isospectral smooth domains, are they isometric?

Open Problem 2.24. If $\Omega_1, \Omega_2 \subset \mathbb{R}^2$ are isospectral convex domains, are they isometric?

2.3 Spectral invariants

Because of all these counterexamples, the goal of hearing the shape of a drum was infeasible. The object of inverse spectral geometry now is to recognize what information we *can* pin down about a manifold from its spectrum. From Weyls' law (Theorem 2.5), we know that volume is a spectral invariant. We can even compute it just from the eigenvalues as $\mathrm{Vol}(M) = \frac{1}{C_n} \lim_{\lambda \to \infty} \lambda^{-\frac{n}{2}} N_k(\lambda)$. In this section, we will describe more such invariants, called the heat invariants and wave invariants, which depend only on the eigenvalues.

2.3.1 Heat invariants

Let (M,g) be a closed Riemannian manifold and let $\lambda_1 \le \lambda_2 \le \dots$ be its Laplace eigenvalues. Suppose we want to solve the heat equation $u_t + \Delta u = 0$ given the initial heat distribution $u(x,0) = u_0(x)$. Since this is a linear equation, we could ask, "how would heat spread if all the initial heat was concentrated at some x"? Then, we could add together all the solutions. Because there are infinitely many points, instead of adding them together, we take an integral. This idea is made precise by the *heat kernel*.

Definition 2.25. The heat kernel $H : \mathbb{R}_{>0} \times M \times M \to \mathbb{R}$ is a function such that for fixed x, the $u_x(t,y) = H(t,x,y)$ is the solution to the heat equation with $u_0(y) = \delta_x(y)$, the Dirac delta distribution on x. To make this precise, for any smooth f on M,

$$\lim_{t\to 0} \int_M H(t,x,y) f(y)\, dy = f(x)$$

where dy is shorthand for integrating using the Riemannian volume on the y factor.

Remark 2.26. The heat kernel can also be interpreted using Brownian motion, by saying that $H(t,x,y)$ is the probablity that a particle at point x will reach point y after time t following Brownian motion. This interpretation requires it to be positive; both the proof of this and this interpretation is explained in [Ros97, §3.4].

Example 2.27. On familiar $\mathbb{R}^n$ (with the standard metric), the heat kernel is given by

$$H(t,x,y) = \frac{1}{(4\pi)^{\frac{n}{2}}} e^{-\frac{\|x-y\|^2}{4t}}$$

which is a Gaussian centered at x for a fixed x. The interpretation is that if all of the heat was concentrated at a point at $t = 0$, the heat would spread radially according to this Gaussian. In fact, even on an arbitrary manifold, we will see that the short-time behavior of the heat kernel is like $\frac{1}{(4\pi)^{\frac{n}{2}}} e^{-\frac{d(x,y)^2}{4t}}$, and the way it differs from this gives us the heat-invariants. This is true even when there is a boundary, by Kac's "principle of not feeling the boundary," [Kac66] which says that at short enough times the heat has not had enough time to each the boundary (from an interior point) and behaves as if the boundary isn't there.

The important fact about the heat kernel is that integrating it with the initial heat distribution tell us how that initial distribution evolves. We only have to calculate the heat kernel once per manifold, and then we know how any initial heat distribution will evolve. The heat kernel exists for any compact Riemannian manifold. For a proof, see [LMP23]; the construction uses the Euclidean heat kernel and perturbs it.

Proposition 2.28 ([LMP23, Theorem 6.1.2])**.** On a closed Riemannian manifold (M,g), there exists a unique heat kernel $H(t,x,y)$, given by $H(t,x,y) = \sum_{k=1}^{\infty} e^{-\lambda_k t} u_j(x) u_j(y)$ where $\{u_i\}$ is an orthonormal Laplace eigenbasis of L^2. The series here converges pointwise on $\mathbb{R}_+ \times M \times M$

This unique heat kernel lets us solve the heat equation for an arbitrary initial heat distribution.

Proposition 2.29. For any initial heat distribution $u_0(x)$, the solution to the heat equation with $u(0,x) = u_0(x)$ is given by $u(t,y) = \int_M H(t,x,y) u_0(x) dx$.

Proof sketch. Let u_0 be an initial heat distribution. Then

$$u(t,y) = \int_M H(t,x,y) u_0(x)\, dx = \int_M \sum_{j=1}^{\infty} e^{-\lambda_j t} u_j(x) u_j(y) u_0(x)$$

Then,

$$\frac{\partial u}{\partial t} = \sum_{j=1}^{\infty} -\lambda_j e^{-\lambda_j t} \int_M u_j(x) u_j(y) u_0(x)\, dx$$

and $\Delta_y u = \sum_{j=1}^{\infty} e^{-\lambda_j t} \int_M u_j(x)(\lambda_j u_j(y)) u_0(x)$ (recall that we are integrating along the x-factor but the heat equation is along the y-factor, so all we need is $\Delta u_j = \lambda_j u_j$). This satisfies the heat equation $\frac{\partial u}{\partial t} + \Delta u = 0$. □

Motivated by the form of the heat kernel, we define the heat trace.

Definition 2.30. The *heat trace* of a manifold is the function $f(t) = \sum_{k=0}^{\infty} e^{-\lambda_k t}$.

The heat trace converges for all $t > 0$ since $f(t) = \sum_{k=1}^{\infty} e^{-\lambda_k t} \int_M u_j^2(x)dx = \int_M H(t,x,x)dx$, where we have used the fact that the u_j can be chosen to be L^2 orthonormal. Minakshisundaram and Pleijel proved the following characterization of the short-time behavior of the heat kernel:

Theorem 2.31 ([LMP23, Theorem 6.1.8], Minakshisundaram-Pleijel expansion)**.** On a closed n-dimensional Riemannia manifold, as $t \to 0^+$,

$$H(t,x,x) \simeq (4\pi t)^{-\frac{t}{2}} \left(\sum_{j=0}^{\infty} a_j(x) t^j \right)$$

where the $a_j(x)$ are called the *local heat invariants* and depend only on the local geometry around x

Now, using the fact that $\sum_{k=0}^{\infty} e^{-\lambda_k t} = \int_M H(t,x,x)dx$, we get the heat-trace expansion

$$\sum_{k=0}^{\infty} e^{-\lambda_k t} = (4\pi t)^{-\frac{t}{2}} \sum_{j=0}^{\infty} a_j t^j$$

where $a_j = \int_M a_j(x)dx$ are the *heat invariants* (also called heat coefficients). The local heat invariants can be computed as expressions in the curvature, its contractions, and covariant derivatives ([Ros97, §3.3], [BGM71]). The first few of these are,

$$a_0(x) = 1, \qquad a_1(x) = \frac{1}{6}\mathrm{Scal}(x), \qquad a_2(x) = \frac{1}{360}(2|\mathrm{Riem}|^2 - 2|\mathrm{Ric}(x)|^2 + 5|\mathrm{Scal}(x)|^2)$$

As we can already start to see, the expressions for these local heat invariants get more and more unwieldy as we increase the order. In general it is quite hard to extract information from the heat invariants beyond the first three or four. However, even these can be used to prove that the spectrum encodes topology for surfaces.

Theorem 2.32. If M and N are two isospectral closed orientable surfaces, then they are homeomorphic.

Proof. For a surface, $\mathrm{Scal}(x) = K(x)$, the Gaussian curvature at x. By the Gauss-Bonnet theorem,

$$a_1 = \int_M \mathrm{Scal}(x)dx = \int_M K(x)dx = 2\pi\chi(M) = 4\pi(1-g)$$

where $\chi(M)$ is the Euler characteristic and g is the genus. Thus $g = 1 - \frac{1}{4\pi}a_1$ and is determined by the spectrum. Then $g(M) = g(N)$ (and we assumed they are both orientable), so M is homeomorphic to N. □

Unfortunately, it is unknown if we can hear orientability for closed surfaces [BW21, §7]. As for determining the surface up to isometry, instead of homeomorphism, the problem is wide open for sphere and tori. However, if we restrict to metrics of constant curvature, then we have positive results for spheres and tori. Isospectral round spheres are isometric by work of Tanno [Tan73] (in fact this holds for dimension up to 5), which also says that a sphere isospectral to a round sphere must be round. As we saw in Section 2.2.1, two isospectral flat 2-tori are isometric (and it holds up to dimension 3), and by using $a_2(x) = \int_M K^2(x)$ we can tell that a torus isospectral to a flat one must itself be flat. However, there are many examples of isospectral but not isometric hyperbolic surfaces, i.e. surfaces with constant curvature -1, such as those found by Vignéras [Vig80]. Buser [Bus92] shows how to adapt the Sunada method to construct many such surfaces.

Corollary 2.33. If M and N are isospectral surfaces of constant sectional curvature, they are locally isometric

Weyl's Law for a closed manifold (Theorem 2.5) can be proved using the fact that $a_0 = \mathrm{Vol}(M)$ and using the Hardy-Little-Karamata Tauberian theoerem, see [LMP23, Theorem 6.1.9]. One drawback to this (as opposed to variational methods) is that it gives weaker remainder estimates. The heat trace expansion can be extended to manifolds with boundary, which in the two dimensional case gets us the following result of Kac [Kac66] (also [LMP23, Theorem 6.1.14]):

Theorem 2.34. Let $\Omega \subset \mathbb{R}^2$ be a planar domain with r smooth boundary components (i.e. an annulus has $r = 2$). Then the heat trace (with Dirichlet eigenvalues) satisfies

$$\sum_{k=1}^{\infty} e^{-\lambda_k t} = \frac{\mathrm{Area}(\Omega)}{4\pi t} - \frac{\mathrm{Length}(\partial\Omega)}{8\sqrt{\pi t}} + \frac{(2-r)}{6} + o(1)$$

This can be modified if there are corners, but those modifications are more complicated and depend on the angles at the corners. Because of the $\frac{(2-r)}{6}$ term, we can hear the number of holes in a smooth planar drum.

Since the heat invariants depend on local geometry, they are good candidates for distinguishing between manifolds with different model geometries. These are the higher dimensional analogues of the spherical, elliptic, and hyperbolic classification in dimension 2, and as we saw with Corollary 2.33, we can 'hear' the local geometry in dimension 2. In dimension 3, the natural analogues are the Thurston geometries, defined in Section 3, and in Section 4, we show that we can recover the model geometry of a (geometric) 3-manifold from its spectrum.

However, this is also the drawback of heat invariants. Because they depend on local geometry, they cannot tell apart locally isometric spaces. There are a variety of tools that can be used in this case, and the main one we will focus on are the wave invariants.

2.3.2 The wave trace and the length spectrum

Much like how the heat equation gives us the heat trace, which gives us the heat invariants, we can also recover spectral information from the wave equation. However, the situation here is a little more complicated, because the wave trace is not a function but a tempered distribution.

Definition 2.35. Let (M, g) be a compact Riemannian manifold (with or without boundary). Then

$$w(t) = \sum_{k=1}^{\infty} e^{-i\sqrt{\lambda_k} t}$$

is called the *wave trace*. Its singular support $\mathrm{SingSupp}(w(t))$ is the complement of the largest open set on which $w(t)$ is smooth.

The most important fact about the wave trace is that it is singular only at lengths of closed geodesics. This fact is called the Poisson relation, and it was proven in different generalities by Chazarain [Cha74], Colin de Verdière [CdV73a] [CdV73b], and Duistermaat-Guillemin [DG75].

Theorem 2.36 (Poisson relation)**.** Let (M, g) be a closed Riemannian manifold, $w(t)$ its wave trace, and $\mathrm{Spec}_L \pm(M)$ be the length spectrum i.e. the set of lengths of closed geodesics. Let $\mathrm{Spec}_L^{\pm}(M) = \{\pm l \mid l \in \mathrm{Spec}_L(M)\}$. Then,

$$\mathrm{SingSupp}(w(t)) \subseteq \mathrm{Spec}_L(M)$$

The Poisson relation is cleanest to state for compact manifolds without boundary. Otherwise, we have to define what a geodesic does at the boundary; see Remark 2.11. For a smooth boundary, a geodesic reflects by keeping the angle of incident equal to the angle of reflection, giving us the *periodic billiards*.

If the Poisson relation was an equality, then the wave invariants would provide us a way to recover geometric data from spectral data in the following way. Take two isospectral spaces, and assume that the Poisson relation

is an equality. This means that they must have the same lengths of closed geodesics. Using this, we can try to recover the space itself. Using the billiard flow, Lu and Rowlett in the expository article [LR15] show how one can determine certain trapezoids from there length spectrum (they show in [HLR17] that certain trapezoidal lengths are spectral invariants using a trace formula, bypassing the need to establish that the Poisson relation is an equality).

In certain special cases, we can explicitly calculate the spectrum or the geodesics relate the Laplace spectrum and the length spectrum. Poisson's summation formula (Theorem 2.13) is one such example, and from it we deduce that isospectral flat tori have the same length spectrum (Theorem 2.15). Similarly, for hyperbolic surfaces, we have the Selberg trace formula. For the precise conditions on h, see the reference given. A primitive geodesic is a closed geodesic that is traversed only once.

Theorem 2.37 (Selberg Trace formula, [Bus92, §9.5.1])**.** Let M be a compact hyperbolic surface. Let h be a holomorphic even function on the upper half plane with certain decay properties, and let g be its Fourier transform. Let γ be a closed geodesic and $l(\gamma)$ be its length. Let γ_0 be the unique primitive geodesic of which γ is a multiple. Then

$$\sum_{k=1}^{\infty} h(\lambda_k) = \frac{\text{Area}(M)}{4\pi}\int_{-\infty}^{\infty} rh(r)\tanh(\pi r)\,dr + \sum_{\gamma}\frac{l(\gamma_0)g(l(\gamma)}{2\sinh(l(\gamma)/2)}$$

From the Selberg trace formula, one can deduce Huber's theorem

Theorem 2.38 (Huber's theoerem, [Bus92, Thm 9.])**.** Two compact hyperbolic surfaces are isospectral if and only if they have the same length spectrum

Thus, in all these special cases, the Laplace spectrum completely determines the length spectrum, allowing us to translate spectral information into geometric information. If the Poisson relation is and equality for a class of Riemannian manifolds, then within that class, isospectrality would imply having the same length spectrum.

Unfortunately, we do not know whether the Poisson relation is always an equality, not do we have an example in which it is not. In [DG75], Duistermaat and Guillemin show that for a bumpy metric, i.e. a metric where every $l \in \text{Spec}_L(M)$ has a unique geodesic, the Poisson relation is an equality. Sutton [Sut20] shows that for generic bi-invariant metrics on compact Lie groups (which are not bumpy), the Poisson relation is an equality. However, giving evidence in the other direction, Koval and Vig in [KV24] construct certain domains in $\mathbb{R}^2$ where the wave trace can be made arbitrary C^k in the neighborhood of a geodesic length.

To understand why the Poisson relation is not guaranteed to be an equality, we need to look at the Duistermaat-Guillemin trace formula. This formula involves certain coefficients, the *wave invariants*, which arise from Fourier transforms of the wave trace. These are expressed as sums over fixed-point sets of the geodesic flow, and the main worry is that these sums could cancel, smoothing out the singularity at a certain geodesic length. To make this precise, we need certain definitions.

Definition 2.39. Let $\tau \in \text{Spec}_L(M)$. Let Φ_τ be the time τ geodesic flow, which is the flow on the unit tangent bundle SM given by $v \mapsto d\gamma_v(\tau)$ where γ_v is the unique geodesic in the direction of v. Denote $\text{Fix}(\Phi_\tau)$ be the fixed-point set of the time τ geodesic flow, which corresponds to closed geodesics of length τ. A length τ is said to be *clean* if τ is a non-degenerate value of the energy functional $E\colon \Omega(M,g) \to \mathbb{R}$

If τ is a clean geodesic, then its fixed-point $\text{Fix}(\Phi_\tau)$ set consists of finitely many disjoint manifolds. Suppose τ is a clean geodesic. Then Duistermaat-Guillemin show that in an interval I_τ containing τ and no other closed geodesic, there exist compactly supported distributions β^{even} and β^{odd} which are only singular at 0, such that $w(t) = \beta^{\text{even}}(t-\tau) + \beta^{\text{odd}}(t-\tau)$ on I_τ.

Let $\bullet$ denote odd or even. Each distribution $\beta^\bullet$ has a smooth Fourier transform $\alpha^\bullet$, and there exist complex numbers $\text{Wave}_k^\bullet(\tau)$ such that asymptotically as $s \to \infty$

$$\alpha^\bullet(s) = \sum_{k=0}^{\infty} \text{Wave}_k^\bullet(\tau) s^{(D_\bullet - 2k-1)/2}$$

where D_{even} (resp odd) is the maximum even dimension of a component of $\text{Fix}(\Phi_\tau)$. These numbers $\text{Wave}_k^\bullet(\tau)$ are called the wave invariants. If all the wave invariants at τ are zero, then $w(t)$ is not singular at τ. For a clean period τ, Duistermaat-Guillemin [DG75, Theorem 4.5] give a formula for the 0th wave invariant. Let $\Theta_1, \dots, \Theta_k$ be the components of $\text{Fix}(\Phi_\tau)$. Then,

$$\text{Wave}_0^\bullet(\tau) = \left(\frac{1}{2\pi i}\right)^{\frac{D_\bullet - 1}{2}} \sum_{j=1, \dim \Theta_j = D_\bullet} i^{-\sigma_j} \int_{\Theta_j} d\mu_j^\tau$$

Here, σ_j is the Morse index of a closed geodesic of length τ with initial velocity in Θ_j, and μ_j^τ is a canonical positive measure on Θ_j called the Duistermaat–Guillemin measure [Sut20]. To show that a geodesic length τ is audible, i.e. produces a singularity of the wave trace, we need to show that the length is clean and at least one of the wave invariants is nonzero. For the 0th wave invariant, if there are multiple components Θ_j of the fixed point set, there is a possibility that their contributions to the sum cancel, leaving us 0. If the metric is bumpy, then there is only one point in $\text{Fix}(\Phi_\tau)$ corresponding to the unique closed geodesic of that length, and so the 0th wave invariant is always nonzero. Thus, for bumpy metrics, the Poisson relation is an equality.

Remark 2.40. Duistermaat-Guillemin's results in [DG75] work for any elliptic operator. The closed geodesics are replaced by periodic orbits of the Hamiltonian flow given by its principal symbol, and the Morse index is replaced by the Maslov index. These general results can be applied to the operator $\sqrt{\Delta}$. Then, as Duistermaat shows in [Dui76], for a Riemannian metric the Maslov index of a periodic orbit of the (co)geodesic flow (on the unit *cotangent* bundle S^*M) is the classical Morse index of the corresponding geodesic.

To make use of the wave invariants, one doesn't need to show that the entire length spectrum is audible. For instance, Hezari-Zeldtich show that a certain class of geodesics are always audible in domains isospectral to ellipses of small eccentricity by showing the wave trace is always singular at those geodesics ([HZ22, §6]). Using this, they prove that one can hear the shape of ellipses of small eccentricity:

Theorem 2.41. There exists an $\epsilon_0 > 0$ such that if $\Omega \subset \mathbb{R}^2$ is (Dirichlet or Neumann) isospectral to an ellipse of eccentricity less than ϵ_0, then Ω is isometric to that ellipse.

We will use the wave invariants in Section 5 to prove that certain classes of Berger lens spaces are not isospectral. Heat invariants cannot be used for this purpose since these spaces are locally isometric. By relying on geodesics, wave invariants capture more global data. However, thee invariants are not entirely global. As Zelditch remarks in [Zel04], they are global in the sense that they depend on geodesics, but they are local in the sense that they only depend on the metric in a small neighborhood of a geodesic. If we changed the metric away from all geosesics of length τ, we would not change the wave invariants at τ at all. This only applies to $\tau \neq 0$; since $\text{Fix}(\Phi_0)$ is all of SM, the wave invariants at $\tau = 0$ are essentially equivalent to the heat invariants [Zel04, §4.1].

3 Geometric structures on Three-Manifolds

Traditionally, spectral geometry is often done on manifolds without much symmetry, or with "bumpy" metrics, to get generic results. In this thesis we will look at spectral geometry in highly symmetric contexts, following Sutton's program from [Sut21], [LSS24], and [LSS21]. The most highly symmetric geometries are those with constant sectional curvature. For dimension ≤ 5, Tanno [Tan73] showed that a manifold isospectral to one with constant sectional curvature must have constant sectional curvature. We saw the two-dimensinal case of this in Section 2.3. To generalize a little more, we look at the three-dimensional Thurston geometries. These are the three-dimensional analogues of the classification of two-dimensional geometries into Euclidean, elliptic, and hyperbolic. In three dimensions, the Thurston geometries provide a paradigm that is highly symmetric, while general enough to apply to every manifold (via the Geometrization theorem).

3.1 Geometric manifolds

Intuitively, a "geometry" is a sort of "fabric" that you can make a manifold out of. In two dimensions, there are three: Euclidean, spherical (or elliptical), and hyperbolic. Manifolds which are "made out of" these fabrics are highly symmetric, because their local properties depend only on the fabric. We shall now proceed to make these intuitions precise.

Definition 3.1. A Riemannian manifold (M,g) is *locally homogeneous* if for every pair of points $p,q \in M$ there exist neighborhoods $U \ni p, V \ni q$ such that U is isometric to V. A Riemannian manifold is called *homogeneous* if its isometry group acts transitively on it, i.e. for all pairs $p,q \in M$ there is an isometry $\varphi \in \mathrm{Isom}(M,g)$ such that $\varphi(p) = q$

A homogeneous manifold is locally homogeneous because any neighborhood $U \ni p$ is isometric to $\varphi(U) \ni \varphi(p) = q$. The converse is not true; for instance, the Berger lens spaces we will study in Section 5 is locally homogeneous but not homogeneous. A locally homogeneous manifold may look the same everywhere, but its isometry group is not large enough to carry every point to every other point.

Theorem 3.2 (Singer, [Sin60])**.** A simply-connected locally homogenenous space is homogeneous. Hence the universal Riemannian cover of a locally homogeneous space is homogeneous.

In Thurston's terminology, the universal cover of a locally homogeneous manifold (M,g) together with its group of isometries is the *geometry*, and M is said to be *modeled after* that geometry.

Definition 3.3. A *geometry* is a tuple (X,G) where X is a connected simply-connected manifold, G is a connected Lie group acting smoothly and transitively on X via diffeomorphisms. Moreover, X must admit at least one G-invariant metric and at least one compact quotient $\Gamma\backslash X$ where Γ is a discrete subgroup of G.

Denote by $\mathscr{R}_G(X)$ the set of G-invariant Riemannian metrics on X. For any $g \in \mathscr{R}_G(X)$, the Riemannian manifold (M,g) is homogeneous since $G \le \mathrm{Isom}(M,g)$ acts transitively on M. Thus $\mathscr{R}_G(X) \subseteq \mathscr{R}_{\mathrm{hom}}(X)$, the set of all homogeneous metrics on X. The condition that this is nonempty is equivalent to G acting with compact point-stabilizers G_p:

Proposition 3.4. Let X be a homogeneous space of G, with G acting effectively. Then $\mathscr{R}_G(X)$ is nonempty if and only if the stabilizer subgroup $G_p = \{g \in G \mid g \cdot p = p\}$

Proof. Suppose X admits an invariant metric g. Since an isometry $\varphi\colon X \to X$ fixing a point p is determined uniquely by $d\varphi_p$ (using the Riemannian exponential map), G_p admits a faithful representation into $O(T_pX)$, which is compact. Conversely, suppose that G_p is compact, in which case it admits a Haar measure $d\phi$. Choose a point $p \in X$ and any metric $\langle\cdot,\cdot\rangle$ on T_pX. Define $g_p(v,w) \int_{G_p} g(d\phi v, d\phi w) d\phi$ so that for any $\psi \in G_p$, $\phi^* g_p = g_p$. For any other point q choose any isometry ψ taking q to p and define $g_q(v,w) = g_p(d\psi v, \psi w)$; for any other isometry ψ' we have $\psi \circ (\psi')^{-1} \in G_p$ so this definition does not depend on the isometry chosen. By construction, this is G-invariant. □

Definition 3.5. Let (M,g) be a Riemannian manifold. We say it has a *geometric structure modeled after* (X,G) if the Riemannian universal cover of M is isometric to X equipped with a G-invariant metric.

If $(\widetilde{M},\widetilde{g})$ is the universal Riemannian cover of (M,g), then (M,g) is modeled after $(\widetilde{M},G)$ for any $G \le \mathrm{Isom}(\widetilde{M},\widetilde{g})$ that acts transitively on $(\widetilde{M},\widetilde{g})$.

Proposition 3.6. A Riemannian manifold with a geometric structure is locally homogeneous, and every compact locally homogeneous manifold is modeled after some geometry.

Proof. If (M,g) is modeled after a geometry, its universal cover is homogeneous, and (M,g) is locally isometric to its universal cover. Thus for every $p,q \in M$, there exist lifts $\widetilde{p},\widetilde{q}$ with isometric neighborhoods $\widetilde{U} \ni \widetilde{p}, \widetilde{V} \ni \widetilde{q}$. Then the projections of those neighborhoods are the required isometric neighborhoods of p and q.

In the other direction, the universal cover of a locally homogeneous manifold (M,g) is a simply connected homogeneous manifold $(\widetilde{M},\widetilde{g})$, so (M,g) is modeled after $(\widetilde{M},\mathrm{Isom}(\widetilde{M},\widetilde{g})^o)$ (where we take the identity component of the isometry group to keep the group in the definition of a geometry connected). □

Definition 3.7. From the above, we define a *geometric structure* on a manifold M to be a complete locally-homogeneous metric on M. A manifold M is said to be *geometric* if it admits a geometric structure

Two geometries (X,G) and X',G' are said to be *equivalent* if there is a diffeomorphism $\Phi\colon X\to X'$ and a Lie group isomorphism $\Psi\colon G\to G'$ that intertwines the actions of G on X and G' on X' (i.e. $\Psi(g)\cdot\Phi(x)=\Phi(g\cdot x)$). A geometry (X,G) is said to be a *subgeometry* of X',G' if $X=X'$ and $G\leq G'$ and the action of G' on X restricts to that of G.

Definition 3.8. A geometry (X,G) is said to be *symmetry maximal* if it is not a proper subgeometry of any geometry.

Remark 3.9. In Thurston's terminology, this condition is often just written as *maximal* instead of *symmetry maximal*, but we use the longer term to avoid confusing it with *metrically maximal*, which we define below.

Example 3.10. Consider the round sphere $(S^n,\mathring{g}^n)$. This is the metric induced on S^n by pulling back along the inclusion of S^n into $\mathbb{R}^{n+1}$, where the latter is endowed with its standard metric. Its isometry group is $O(n+1)$ whose identity component is $SO(n+1)$. Thus, $(S^n,\mathring{g}^n)$ is modeled after the geometry $(S^n,SO(n+1))$. In fact, since $SO(n+1)$ has dimension $\frac{n(n+1)}{2}$, this geometry is maximal. By the orbit-stabilizer theorem from Lie groups, any G that acts transitively on a manifold X satisfies $\dim G=\dim X+\dim G_p$, where G_p is the stabilizer subgroup [Lee12, Theorem 21.20]. As we saw in the proof of Proposition 3.4, G_p admits a faithful representation into $O(T_pX)$ and thus cannot have dimension greater than $\frac{n(n-1)}{2}$. Then $\dim G\leq n+\frac{n(n-1)}{2}=\frac{n(n+1)}{2}$. Because we require G to be connected, $SO(n+1)$ cannot be a proper subgeometry of any larger geometry, so $(S^n,\mathring{g}^n)$ is symmetry maximal geometry.

If $(X,H)\leq(X,G)$ then $\mathscr{R}_G(X)\subseteq\mathscr{R}_H(X)$, since an H-invariant metric is also G-invariant. Thus, the symmetry maximal geometries are the ones that have the fewest invariant metrics. For $(S^n,SO(n+1))$ they are just the scaled versions of the round metric. To get all the locally homogeneous metrics on X, we instead look to the concept of *metrically maximal geometries*, which have as many invariant metrics as possible while still being transitive, and maximal with respect to this.

Definition 3.11 ([LSS24])**.** A geometry (X,G) is said to be metrically maximal whenever $(X,H)\leq(X,G)\lneq(X,L)$ implies $\mathscr{R}_L(X)\subsetneq\mathscr{R}_G(X)\subseteq R_H(X)$. That is, shrinking G does not give us any more metrics, but enlarging G gives us strictly fewer metrics.

Example 3.12. By [Zil82], the space of homogeneous metrics on S^3 is $\mathscr{R}_{Sp(1)}(S^3)=\mathscr{R}_{S^3}(S^3)$ where S^3 is viewed as the Lie group of unit quaternions. This is the space of all left-invariant metrics on S^3. Any subgroup of S^3 does not act transitively on it, and it can be shown that enlarging the group will lose some metrics. Thus, (S^3,S^3) is a metrically maximal geometry. Since $S^3\leq SO(4)$, this metrically maximal geometry on S^3 is a subgeometry of the symmetry maximal geometry $(S^3,SO(4))$.

In three-dimensions, it turns out that there is precisely one metrically maximal subgeometry of a symmetry maximal geometry. As we will see in the next section, every geometric M can be modeled on a symmetry maximal geometry, and conversely every locally homogeneous metric on such an M is modeled after a metrically maximal geometry.

3.2 The three-dimensional geometries

In this section we describe the three-dimensional symmetry- and metrically-maximal geometries. There are eight of each, and the symmetry-maximal 3D geometries are called *Thurston geometries*.

3.2.1 The symmetry maximal geometries

Theorem 3.13 (Thurston's Geometrization theorem)**.** Every closed three-dimensional manifold can be cut along embedded spheres and tori into geometric pieces, each of which admits exactly one of eight symmetry-maximal geometries.

This theorem was conjectured by Thurston in the 80s [Thu82], and its proof was completed by Perelman's work on Ricci flow with surgery, which also proves the Poincaré conjecture, in 2002–2003 [Per02] [Per03b] [Per03a]. The difficult part is proving that the pieces so obtained admit geometric structures. We will outline the easier parts of the proof, namely that there are just eight symmetry maximal geometries, and that a geometric manifold admits a unique one.

First, we need to describe the eight geometries. The constant curvature spaces $\mathbb{E}^3, \mathbb{H}^3$, and S^3 are the three-dimensional Euclidean, spherical, and hyperbolic spaces. The product geometries $S^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$ have the spherical or hyperbolic geometry on the left factor and the standard Euclidean geometry on the right factor. The underlying spaces of the remaining three geometries are all $\mathbb{R}^3$, which are all Lie groups. First, we have $\widetilde{SL_2}(\mathbb{R})$, the universal cover of $\mathrm{SL}_2(R)$, thought of as a Lie group. Then, we have the Nilgeometry Nil, which is the group of 3×3 unit upper-triangular matrices; the space is so named because this is a nilpotent group. Solvegeometry Sol is the geometry of the solvable group $\mathbb{R}^2 \ltimes \mathbb{R}$ where $\mathbb{R}$ acts on $\mathbb{R}^2$ via $t \cdot (x, y) = (e^t x, e^{-t} y)$.

Theorem 3.14. If (X, G) is a three dimensional symmetry maximal geometry, X is one of the eight above, and G is chosen according to the Euclidean, hyperbolic spherical, or product isometry groups for the first five, G_{Nil} is generated by Nil and an S^1 action, $G_{\widetilde{SL_2}}$ is generated by $\widetilde{SL_2}$ and an $\mathbb{R}$-action, and G_{Sol} is Sol. All of these are taken to act via left-actions.

The detailed proof of this theorem is given in [Thu97]. Here, we give an outline following [San09, §2.2].

Outline. Let (X, G) be a symmetry maximal geometry. Let G_x be the stabilizer of a point; by homogeneity, the point itself does not matter. As we saw in Example 3.10, G_x is a connected closed subgroup of $SO(3)$. These are just $SO(3), SO(2) \cong S^1$, and I. This gives us three cases.

If $G_x \cong SO(3)$ then the metric is isotropic. Now, take any two planes $\Pi, \Pi' \subseteq T_x X$ and let v, v' be unit vectors perpendicular to Π, Π'. Then, any isometry taking v to v' must take Π to Π'. We can find such an isometry since the metric is isotropic, s we can take any plane to any other. Thus, the metric has constant sectional curvature and $X = \mathbb{E}^3, \mathbb{H}^3, S^3$ with the corresponding isometry group.

If $G_x \cong SO(2)$, let $L_x \subseteq T_x X$ be the invariant subspace of the action and P_x be its orthogonal complement. The action of G must carry L_x to L_y, so we get a vector field V and a plane field P. The flow ϕ_t of the vector field preserves the plane field. If this plane field is integrable, then we get a foliation of the manifold and get one of the product geometries. We discard $\mathbb{E}^2 \times \mathbb{R}$ because it has already been accounted for, giving us two new geometries.

If $G_x \cong I$ then X is diffeomorphic to G_e, the connected component of the identity of G. Since (X, G) admits finite quotients, there is a discrete subgroup Γ such that X/Γ is compact. By Lemma 3.16, G_e must be unimodular, and since it is diffeomeorphic to X it must be simply connected. Milnor in [Mil76] showed that the three-dimensional simply connected unimodular Lie groups are exactly $S^3, \mathbb{E}^3, \widetilde{\mathrm{Isom}(\mathbb{E}^2)}, \widetilde{SL_2}(\mathbb{R})$, Nil and Sol. Since these groups act on themselves via left-multiplication, $S^3, \mathbb{E}^3, \widetilde{SL_2}(\mathbb{R})$ and Nil are not symmetry maximal as they sit under the obvious symmetry maximal geometries. Similarly, $\widetilde{\mathrm{Isom}(\mathbb{E}^2)}$ sits under $\mathbb{E}^3$. The only possible candidate here is (Sol, Sol), and Milnor [Mil76] shows that this is indeed symmetry maximal. □

For this proof, we used the following concept.

Definition 3.15. A Lie group G is called *unimodular* if its left-invariant Haar measure is also right-invariant (or vice-versa)

Lemma 3.16. A Lie group G admitting a discrete subgroup Γ such that G/Γ is compact is unimodular.

Proof sketch. Equip G with a right-invariant Haar measure, which then descends to a measure μ on G/Γ since it is invariant under the deck action. Now, G acts on G/Γ from the left. Define f to be a function such that $(L_g)_*\mu = f(g)\mu$. Since $\int_{G/\gamma}\mu = \int_{G/\Gamma}(L_g)_*\mu$, $f \equiv 1$. Thus μ is invariant under the left-action of G. This lifts to a left- and right-invariant measure on G. □

If M is a manifold admitting a Thurston geometry, it is of the form X/Γ where (X,G) is a symmetry maximal geometry and Γ is a subgroup of G. Because the different geometries have different cocompact discrete subgroups, we can deduce the following theorem.

Theorem 3.17 ([Thu97, Thm 4.7.8])**.** If M is a geometric 3-manifold, it admits a unique Thurston geometry.

Thus, the data of which geometry M might admit is encoded in $\pi_1(M)$. For instance, every 3-manifold with a finite fundamental group which admits a geometry admits an elliptic geometry (modeled after $(S^3, SO(4))$) and no other Thurston geometry.

3.2.2 The metrically maximal geometries

The main drawback of using symmetry-maximal geometries is that non every locally homogeneous manifold is modeled after one. If (M,g) is a locally homogeneous metric, then M admits some metric which is modeled after a symmetry maximal geometry, and this geometry is unique. But g is not guaranteed to be that metric. By looking at geometries with large numbers of invariant metrics, we can try to find a geometry that (M,g) is modeled after. These are the symmetry maximal geometries, and it turns out that in three dimensions each symmetry maximal geometry is a supergeometry of exactly one metrically maximal geometry. Thus, there is a unique metrically maximal geometry that (M,g) is modeled after.

Theorem 3.18 ([LSS24, §1.1])**.** There are eight metrically-maximal geometries: $\widetilde{\mathrm{Isom}(\mathbb{E}^2)}$, $S^3, \mathbb{H}^3, \mathbb{H}^2\times\mathbb{R}, S^2\times\mathbb{R}, \mathrm{Sol}, \mathrm{Nil}, \widetilde{SL_2(\mathbb{R})}$. Of these, $\mathbb{H}^3, \mathbb{H}^2\times\mathbb{R}, S^2\times\mathbb{R}$ and Sol have the same isometry groups as the Thurston geometries, while all the Lie groups' symmetry groups are themselves, acting on the left.

Example 3.19. Let (M,g) be modeled after (S^3,S^3). Then it has a finite fundamental group, so M admits a metric g' such that (M,g') is modeled after $(S^3,SO(4))$. Conversely, if M admits a metric modeled after $(S^3,SO(4))$, then it is a quotient of S^3. Any locally homogeneous metric on M lifts to a homogeneous metric on S^3, which must be left-invariant by [Zil82] (see Example 3.12). Thus if g is a locally homogeneous metric on M, (M,g) is modeled after (S^3,S^3).

Example 3.20 ([LSS24, §1.1])**.** Denote $E(2) = \mathrm{Isom}(\mathbb{E}^2)$. The Lie group $\widetilde{E(2)} = \mathbb{R}^2 \ltimes \mathbb{R}$ where the first factor is translation and the second factor is rotation by θ. There is a map $\Psi : \widetilde{E(2)} \to \mathrm{Isom}(\mathbb{E}^3) = \mathbb{R}^3 \ltimes O(3)$ given by

$$(v,\theta) \mapsto \left((v,\theta), \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}\right)$$

This is a Lie group embedding, and we can check that $\Psi(\widetilde{E(2)})$ acts transitively on $\mathbb{R}^3$ and that $(\widetilde{E(2)}, \widetilde{E(2)}) \cong (\mathbb{R}^3, \Psi(\widetilde{E(2)}))$. So $(\widetilde{E(2)}, \widetilde{E(2)})$ is a subgeometry of $\mathbb{E}^3$. If (M,g) is a locally homogeneous manifold such that M admits a flat metric, then (M,g) is modeled after $(\widetilde{E(2)}, \widetilde{E(2)})$ and vice-versa. Thus $(\widetilde{E(2)}, \widetilde{E(2)})$ is the metrically maximal geometry sitting under $\mathbb{E}^3$.

Proof of Theorem 3.18. Unlike the proof of the Thuston geometries, this proof is essentially an enumeration that assumes a manifold can be given a Thurston geometry and deduces a symmetry group. If (M,g) is locally homogeneous then it is modeled after some geometry (X,G). Either (X,G) is equivalent to a Thurston geometry, or it is a proper subgeometry of one. Sekigawa [Sek77, Theorem B] shows that a proper subgeometry

of Thurston geometry must be (G,G) where G acts on itself via left multiplication. Raymond and Vasquez [RV81] show that the Lie groups that could lead to such geometries are precisely $\mathbb{R}^3, S^3, \mathrm{Nil}, \mathrm{Sol}, \widetilde{SL_2(\mathbb{R})}$, and $\widetilde{E(2)}$. Other than $\mathbb{R}^3$ (which just gives us $\mathbb{E}^3$) and Sol (which gives us the Thurston geometry (Sol, Sol)), the geometries arising from these are subgeometries of a unique Thurston geometry. Thus, if M admits a Thurston geometry based on $S^3, \mathrm{Nil}, \mathrm{Sol}, \widetilde{SL_2(\mathbb{R})}$, any locally homogeneous metric on it is modeled after one of (G,G) for $G = S^3, \mathrm{Nil}, \widetilde{SL_2(\mathbb{R})}$. This finishes the case when the geometry is a proper subgeometry of a Thurston geometry; otherwise, it must be a Thurston geometry $\mathbb{H}^3, \mathbb{H}^2 \times \mathbb{R}, S^2 \times \mathbb{R}$, or (Sol, Sol). □

Corollary 3.21. A compact locally homogeneous 3-manifold (M,g) can be put into one of exactly eight equivalence classes. These can be indexed either by the Thurston geometry that M admits or the metrically-maximal geometry that (M,g) is modeled after.

4 Recovering model geometries from the heat invariants

Outline: Milnor's classification of unimodular Lie groups using Ricci eigenvalues. Lin–Schmidt–Sutton's recovery of Ricci eigenvalues from the heat invariants. Overview of the ambiguous case $\widetilde{SL}_2(\mathbb{R})$ *and* Sol

5 Berger lens spaces and the wave invariants

The three sphere S^3 equipped with a Berger metric covers two non-isometric lens spaces [LSS21]. Exactly one of these spaces is homogeneous, with the other being merely locally homogeneous. It is unknown whether these spaces can be isospectral. We want to prove that they are not.

Our line of attack is to look at the shortest closed geodesic, or closed geodesics in general. Recall from Section 2.3 that for a $\tau \in \mathrm{Spec}_L(M)$, if it is a clean length, we can compute $\mathrm{Wave}_0^{\mathrm{even}}$ and $\mathrm{Wave}_0^{\mathrm{odd}}$; if any of these are nonzero then τ is spectrally determined by virtue of being in $\mathrm{SingSupp}(w(t))$

Suppose M and N were two Riemannian manifolds, such that the lengths of their shortest closed geoedesic was clean, the lengths were different, and we could tell that the lengths induced a singularity in the wave trace. Then we would know that the two manifolds are not isospectral, because the smallest singularity of their wave trace occurs at different times. In fact if there is any τ such that there is a closed geodesic of that length on one manifold but not another, then we need only look at the one and show that it is clean and that the trace formula doesn't cancel.

As we saw in Section 3, a locally homogeneous 3-manifold (M,g) with a finite fundamental group is modeled after (S^3,S^3). In this case M admits a metric modeled after $(S^3,SO(4))$, so M is called a 'spherical' or 'elliptic' 3-manifold. In [LSS21], there is a complete classification of these manifolds. This stems from the fact that if $\widetilde{g}$ is a left-invariant metric on S^3 then $\mathrm{Isom}(S^3,\tilde{g}) \leq \mathrm{Isom}(S^3,\mathring{g}) = O(4)$ [Oni92, Thm 4 and 6] [LSS21, §1.1]. Since $M \cong S^3/\Gamma$ for $\Gamma \leq \mathrm{Isom}(S^3,\tilde{g}) \leq O(4)$, we can enumerate all the diffeomorphism classes of elliptic manifolds by enumerating discrete subgroups of $SO(4)$ acting freely on S^3. Note that S^3/Γ_1 is diffeomorphic to S^3/Γ_2 if and only if Γ_1 and Γ_2 are conjugate in $O(4)$.

The discrete subgroups of $SO(4)$ acting freely on S^3 fall into six infinite families I-VI, of which I contains all the finite cyclic groups. Since we will be concerned with lens spaces, we will only look at Type I groups.

An arbitrary left-invariant metric on S^3 is given by three parameters by choosing an inner product at the origin. Equivalently, we can look at the three eigenvalues of its Gram matrix. If they are all the same, we have the round metric. If they are all different, we have a generic left-invariant metric. If two of them are the same, we have a *Berger metric*. A *Berger sphere* is (S^3,g) with g a Berger metric, and a *Berger lens space* is a lens space covered by a Berger sphere. As discussed above, for $q \geq 3$, a Berger sphere covers exactly two isometry classes of Berger lens spaces, of which at most one is homogeneous (the other is merely locally homogeneous).

We want to show that the two separate Berger lens spaces covered by a Berger sphere are not isospectral by showing that their smallest closed geodesic length is different. The plan to do this is as follows

1. Compute the length of the smallest closed geodesic and see that they are different
2. Show that both of these periods are clean
3. Show that there is no cancellation in the trace formula at these periods

If this holds, then these manifolds are not isospectral. This would rule out a class of counterexamples to the idea that we can heat homogeneity. If cleanliness or lack-of-cancellation fail, then we would have to look at other geodesic lengths.

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